

Chapter 3 Review

Learning Objectives

- Consolidate understanding of database concepts and SQL operations
- Demonstrate proficiency by solving 综合 SQL challenges
- Reflect on real-world applications of database management

Engage (5-7 minutes)

Think back to the first lesson—what a database is and why it matters. Today we connect all the concepts: from designing tables to writing complex queries, every skill builds on the previous one to solve complete database problems.

Explore (10-12 minutes)

Work independently on a comprehensive challenge: given a “School” database with Students, Teachers, Subjects, and Grades tables, write ten SQL queries covering CREATE, INSERT, SELECT with WHERE, UPDATE, DELETE, aggregate functions, GROUP BY, and ORDER BY. Exchange answers with a partner for peer review.

Explain (10-12 minutes)

Review the 5E progression: we Engaged by understanding why databases exist, Explored table structures and relationships, Explained SQL syntax and commands, Elaborated through real-world projects, and Evaluated through quizzes and presentations. Each step deepened our understanding of data management.

Elaborate (8-10 minutes)

Research one real-world database application—hospital records, airline reservations, or online shopping. Explain which tables it might contain, how they relate, and what queries would be most frequently used. Present your findings as a one-page report.

Evaluate (5-8 minutes)

Chapter test: ten questions covering database terminology, SQL syntax for all CRUD operations, aggregate functions, GROUP BY, and WHERE conditions. Score 7 out of 10 to demonstrate mastery. Reflect on which topics need more practice and set goals for the next chapter.